```
function sol = ps_kriging(x,x0,Nmax,dx_max,kriging_method,cv_model)

% Kriging algorithm (for general use)
%
% Input:    - x               data points [cn1 cn2 z]
%           - x0              new points [cn1 cn2]
%           - Nmax            number of nearest neighbors to use
%           - dx_max          search radius for neighbors
%           - kriging_method  kriging method,
%                             1 = Simple Kriging
%                             2 = Ordinary Kriging with one nonbias
%                             constraint
%                             3 = Ordinary Kriging with number of
%                             parameters nonbias constraints
%                             4 = Universal Kriging with drift of
%                             order 1
%                             5 = Universal Kriging with drift of
%                             order 2
%           - cv_model        covariance model
%                             1 = nugget effect
%                             2 = exponential model
%                             3 = gaussian model
%                             4 = spherical model
%                             (new models can easily be added)
%
% Output:   - sol             solution
%
% --------------------------------------------------------------------
% File.............: ps_kriging.m
% Version & Date..: 1.7.2.16, 12-DEC-2009
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% --------------------------------------------------------------------
%
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% intended for scientific use only. Applications for commercial use are
% prohibited.
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%
% Change log
% - v1.7.7.1, normalization of k and k0 by 1000 to avoid singular matrix
%

% -----------------------------------------------------
% Initialize
% -----------------------------------------------------

Nmodel = size(cv_model,1);
Npar = 1;
Nx0 = size(x0,1);
Nx = size(x,1);
Nmax = min(Nmax,Nx);


% -----------------------------------------------------
% Select data points
% -----------------------------------------------------

x0_mean = mean(x0,1);
dx = sqrt((x(:,1)-x0_mean(1)).^2+(x(:,2)-x0_mean(2)).^2);
[dx_sort,index] = sort(dx);
index2 = find(dx_sort(1:Nmax)<dx_max);
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index = index(index2);
x = x(index,:);
Nx = size(x,1);

if Nx<3
  kriging_method=2; %change to ordinary kriging, otherwise singular
                    %k matrix
end

% ------------------------------------------------------
% Setup Kriging system
% ------------------------------------------------------

xa = repmat(x(:,1),1,Nx);
xb = xa';
ya = repmat(x(:,2),1,Nx);
yb = ya';
x0a = repmat(x(:,1),1,Nx0);
x0b = repmat(x0(:,1)',Nx,1);
y0a = repmat(x(:,2),1,Nx0);
y0b = repmat(x0(:,2)',Nx,1);
h = sqrt((xa-xb).^2+(ya-yb).^2);
h0 = sqrt((x0a-x0b).^2+(y0a-y0b).^2);

k = zeros(Nx,Nx);
k0 = zeros(Nx,Nx0);
for w = 1:Nmodel
  k = k+covar_function(h,cv_model(w,:));
  if cv_model(w,1)~=1
    k0 = k0+covar_function(h0,cv_model(w,:));
  end
end

switch kriging_method
  case 1 %Simple Kriging
    avg = mean(x(:,3));
    x(:,3) = x(:,3)-avg;

  case 2 %Ordinary Kriging (one nonbias constraint)
    t = ones(Nx*Npar,1);
    k = [k t;t' 0];
    t = ones(1,Nx0*Npar);
    k0 = [k0;t];

    avg = mean(x(:,3));
    x(:,3) = x(:,3)-avg;

  case 3 %Ordinary Kriging (Npar nonbias constraints)
    error(['Sorry, Kriging with multiple parameters is not ' ...
           'implemented yet']);
    t = kron(ones(Nx,1),eye(Npar));
    k = [k t;t' zeros(Npar)];
    t = kron(ones(1,Nx0),eye(Npar));
    k0 = [k0;t];

  case 4 %Universal Kriging with drift of order 1
    t = kron(ones(Nx,1),eye(Npar));
    t2 = kron(0.001*x(:,1:2),eye(Npar));%x0.001 normalization to
                                        %avoid singular matrix
    k = [k t t2; [t' zeros(Npar,3*Npar)]; [t2' zeros(2*Npar,3*Npar)]];
    t = kron(ones(1,Nx0),eye(Npar));
    t2 = kron(0.001*x0',eye(Npar));;%x0.001 normalization to
                                        %avoid singular matrix
    k0 = [k0;t;t2];
```

```
  case 5 %Universal Kriging with drift of order 2
    error(['Sorry, Universal Kriging with drift of order 2 is ' ...
           'not implemented yet']);

  otherwise
    error('You specified a wrong Kriging method');
end


% ----------------------------------------------------
% Solve the Kriging system
% ----------------------------------------------------

%l = inv(k)*k0;
l = k\k0;
sol = l(1:Nx,:)'*x(:,3);

switch kriging_method
  case 1 %Simple Kriging
    sol = sol+avg;
  case 2 %Ordinary Kriging (one nonbias constraint)
    sol = sol+avg;
end


%subfunction
function cv = covar_function(h,cv_model)

model = cv_model(1);
a = cv_model(2);
c = cv_model(3);

switch model
  case 1 %nugget
    cv = kron(eye(size(h)),c);
  case 2 %exponential
    cv = c*exp(-h/a);
  case 3 %gaussian
    cv = c*exp(-(h/a).^2);
  case 4 %spherical
    cv = c*(1-1.5*min(h/a,1)+0.5*min(h/a,1).^3);
  otherwise
    error('You specified a wrong covariance model');
end
```